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INNER-OUTER NON-SKID SLEEVE WORN ON HAND OR FOOT

Abstract

An inner-outer non-skid sleeve worn on a hand or foot, including a front sleeve portion and a rear sleeve portion coupled together. Inner surfaces of the front sleeve portion and the rear sleeve portion have an inner non-skid layer so as to be partially covered with the inner non-skid layer but partially exposed. The inner non-skid layer has a predetermined thickness and thus protrudes from the inner surfaces of the sleeve body. Outer surfaces of the front sleeve portion and the rear sleeve portion have an outer non-skid layer so as to be partially covered with the outer non-skid layer but partially exposed. The outer non-skid layer is in contact with an external object or the ground. The outer non-skid layer has a predetermined thickness and thus protrudes from the outer surface of the sleeve body.

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Background/Summary

BACKGROUND OF THE INVENTION

1. Field of the Invention

[0001] The present disclosure relates to non-skid hand or foot wearables, and more particularly to an inner-outer non-skid sleeve worn on a hand or foot.

2. Description of Related Art

[0002] Conventional non-skid gloves worn on hands are knitted fabrics fitted around the hands and equipped with non-skid patches or non-skid strips on palmar sides of the gloves to prevent skidding during manual operation. Similarly, conventional non-skid socks worn on feet are knitted fabrics fitted around the feet and equipped with non-skid patches or non-skid strips on plantar sides of the socks to prevent skidding during treading.

[0003] Taiwan patent TW M481614 discloses a non-skid sock with plantar knitted protrusions for preventing skidding and enhancing holding power. Taiwan patent TW M481614 discloses plantar non-skid protrusions but not dorsal non-skid protrusions and thus fails to meet the need for dorsal skidding prevention. Moreover, Taiwan patent TW M481614 does not disclose internal non-skid protrusions and thus fails to meet the need for preventing the foot from sliding relative to the sock.

[0004] Spain patent ES 186368 discloses a non-skid glove with a plurality of non-skid protrusions disposed on the palmar side and a plurality of non-skid protrusions disposed on the dorsal side and thus meets the need for dorsal skidding prevention as well. However, Spain patent ES 186368 does not disclose internal non-skid protrusions either and thus fails to meet the need for preventing the hand from sliding relative to the glove.

[0005] Skidding prevention needs arise from some situations, such as yoga, as people may wear gloves or socks while practicing yoga. Although the prior art enables gloves and socks to have an external non-skid function, the prior art fails to provide any non-skid effect on hands in the internal environment of the gloves and on feet in the internal environment of the socks. Owing to the aforesaid presence of a good external non-skid effect and the aforesaid absence of any internal non-skid effect, it is easy for the hands to slide out of the gloves and the feet to slide out of the socks sometimes.

[0006] Conventional surgical gloves in wide use are made of rubber and have a satisfactory internal non-skid effect on hands. However, when fitted inside conventional surgical gloves and wetted with sweat, surgeons' hands become slippery and sticky to the detriment of the sense of touch during surgery.

[0007] Therefore, it is imperative to address the issues of non-skid functions and human sweat inside gloves and socks.

BRIEF SUMMARY OF THE INVENTION

[0008] It is an objective of the disclosure to provide an inner-outer non-skid sleeve worn on a hand or foot such that, when worn on the human hand or the human foot, the inner-outer non-skid sleeve can absorb sweat and impose a non-skid effect on the outside and the inside.

[0009] Another objective of the disclosure is to provide an inner-outer non-skid sleeve worn on a hand or foot to exhibit a non-skid effect regardless of whether the human hand or the human foot is sweating.

[0010] To achieve the above and other objectives, the disclosure provides an inner-outer non-skid sleeve worn on a hand or foot, comprising: a front sleeve portion and a rear sleeve portion coupled together, the front sleeve portion being fitted to fingers of a human hand or toes of a human foot, and the rear sleeve portion being fitted to a palm of the human hand or a sole of the human foot, wherein the front sleeve portion and the rear sleeve portion are made of a flexible, hygroscopic material and joined to form a sleeve body and an opening, allowing the human hand or the human foot to be inserted into the sleeve body through the opening, wherein surfaces of the front sleeve portion and the rear sleeve portion are defined as inner surfaces when located inside the sleeve body, and the inner surfaces of the front sleeve portion and the rear sleeve portion have an inner

non-skid layer so as to be partially covered with the inner non-skid layer but partially exposed, wherein the human hand or the human foot comes into contact with the inner non-skid layer and the inner surfaces of the front sleeve portion and the rear sleeve portion when the inner-outer non-skid sleeve is worn on the human hand or the human foot to allow the human hand or the human foot to be positioned in the sleeve body, wherein the inner non-skid layer has a predetermined thickness and thus protrudes from the inner surfaces of the sleeve body, wherein surfaces of the front sleeve portion and the rear sleeve portion are defined as outer surfaces when located outside the sleeve body, and the outer surfaces of the front sleeve portion and the rear sleeve portion have an outer non-skid layer so as to be partially covered with the outer non-skid layer but partially exposed, wherein the outer non-skid layer is in contact with a ground when the sleeve body is worn on the human hand or the human foot with the palm pressing against the ground or the human foot treading on the ground, wherein the outer non-skid layer has a predetermined thickness and thus protrudes from the outer surfaces of the sleeve body.

[0011] Therefore, the inner-outer non-skid sleeve worn on the human hand or the human foot absorbs sweat and provides a non-skid effect on the outside and the inside regardless of whether the human hand or the human foot is sweating.

Description

BRIEF DESCRIPTION OF THE SEVERAL VIEWS OF THE DRAWINGS

[0012] FIG. **1** is a perspective view of the first preferred embodiment of the disclosure.

[0013] FIG. **2** is a rear perspective view of the first preferred embodiment of the disclosure.

[0014] FIG. **3** is a dissected schematic view of the first preferred embodiment of the disclosure.

[0015] FIG. **4** is a wearing schematic view of the first preferred embodiment of the disclosure.

[0016] FIG. **5** is a top view of another aspect of the first preferred embodiment of the disclosure, showing its penetrating holes.

[0017] FIG. **6** is a top view of yet another aspect of the first preferred embodiment of the disclosure.

[0018] FIG. **7** is a perspective view of still yet another aspect of the first preferred embodiment of the disclosure.

[0019] FIG. **8** is a perspective view of a further aspect of the first preferred embodiment of the disclosure.

[0020] FIG. **9** is a perspective view of a further aspect of the first preferred embodiment of the disclosure.

[0021] FIG. **10** is a perspective view of a further aspect of the first preferred embodiment of the disclosure.

[0022] FIG. **11** is a top view of a further aspect of the first preferred embodiment of the disclosure.

[0023] FIG. **12** is a perspective view of a further aspect of the first preferred embodiment of the disclosure.

[0024] FIG. **13** is a perspective view of the second preferred embodiment of the disclosure.

[0025] FIG. **14** is a cutaway view of the second preferred embodiment of the disclosure.

DETAILED DESCRIPTION OF THE INVENTION

[0026] The technical features of the disclosure are herein illustrated with preferred embodiments, depicted with drawings, and described below.

[0027] As shown in FIG. **1** through FIG. **4**, the first preferred embodiment of the disclosure provides an inner-outer non-skid sleeve **10** worn on a hand or foot. The inner-outer non-skid sleeve **10** essentially comprises a front sleeve portion **11** and a rear sleeve portion **13**.

[0028] The front sleeve portion **11** and the rear sleeve portion **13** are made of a flexible, hygroscopic material and joined to form a sleeve body **19** and an opening, allowing the human

hand or the human foot to be inserted into the sleeve body **19** through the opening. In the first embodiment, the sleeve body **19** is exemplified by a glove and described herein according to the situation in which it is worn on the human hand. The inner-outer non-skid sleeve **10** is either integrally formed or formed by sewing or adhering its aforesaid constituent elements together. [0029] The front sleeve portion **11** is fitted to fingers of the human hand, and the rear sleeve portion **13** is fitted to the palm of the human hand. In practice, the flexible, hygroscopic material is knitted fabric, Polyurethane (PU) or artificial leather. In this embodiment, the flexible, hygroscopic material is exemplified by knitted fabric, i.e., cloth.

[0030] Furthermore, surfaces of the front sleeve portion **11** and the rear sleeve portion **13** are defined as inner surfaces **191** of the sleeve body **19** when located inside the sleeve body **19**. The inner surfaces **191** of the front sleeve portion **11** and the rear sleeve portion **13** have an inner non-skid layer **21** so as to be partially covered with the inner non-skid layer **21** but partially exposed. When the sleeve body **19** is worn on the human hand, the human hand is inside the sleeve body **19** and thus is in contact with the inner non-skid layer **21** and the inner surfaces **191** of the front sleeve portion **11** and the rear sleeve portion **13**. The inner non-skid layer **21** has a predetermined thickness and thus protrudes from the inner surfaces **191** of the sleeve body **19**.

[0031] Furthermore, surfaces of the front sleeve portion **11** and the rear sleeve portion **13** are defined as outer surfaces **192** of the sleeve body **19** when located outside the sleeve body **19**. The outer surfaces **192** of the front sleeve portion **11** and the rear sleeve portion **13** have an outer non-skid layer **23** so as to be partially covered with the outer non-skid layer **23** but partially exposed. When the sleeve body **19** is worn on the human hand, the palm presses against the ground, allowing the outer non-skid layer **23** to be in contact with the ground. The outer non-skid layer **23** has a predetermined thickness and thus protrudes from the outer surfaces **192** of the sleeve body **19**.

[0032] As shown in FIG. **1** through FIG. **4**, when the sleeve body **19** is worn on the human hand, the outer non-skid layer **23** corresponds in position to the fingers and the palm. As shown in FIG. **3** and FIG. **4**, the inner non-skid layer **21** inside the sleeve body **19** encloses the human hand when the sleeve body **19** is worn on the human hand. Therefore, when the inner-outer non-skid sleeve **10** is in use, for example, during a yoga workout, not only does the outer non-skid layer **23** of the outer surfaces **192** of the sleeve body **19** have a non-skid effect on the ground when the palm presses against the ground, but the inner non-skid layer **21** on the inner surfaces **191** of the sleeve body **19** also has a non-skid effect on the human hand. Furthermore, the sleeve body **19** is made of a hygroscopic material, such as knitted fabric (i.e., cloth), and thus absorbs sweat when the human hand is sweating; thus, the sweat is unlikely to accumulate between the inner non-skid layer **21** and the skin of the human hand, and in consequence the non-skid effect of the inner non-skid layer **21** on the human hand remains unabated. Therefore, the inner non-skid layer **21** has a non-skid effect on the human hand, regardless of whether the human hand is sweating.

[0033] In the first embodiment illustrated by FIG. **1** through FIG. **4**, for example, the inner non-skid layer **21** and the inner surface **191** of the front sleeve portion **11** as well as the outer non-skid layer **23** and the outer surface **192** of the front sleeve portion **11** are necessarily disposed at the front sleeve portion **11** and correspond in position to the fingerprints of the fingers and the ventral sides of the fingers of the human hand. When the human hand manipulates an object or when the palm presses against the ground, the fingerprints of the fingers and the ventral sides of the fingers are usually the points of contact. Thus, inside the sleeve body **19**, the inner non-skid layer **21** and the inner surface **191** of the front sleeve portion **11** are in contact with the fingerprints of the fingers and the ventral sides of the fingers of the human hand. Furthermore, outside the sleeve body **19**, the outer non-skid layer **23** and the outer surface **192** of the front sleeve portion **11** are in contact with the ground when the fingers of the human hand press against the ground, providing a non-skid effect. When the sleeve body **19** is worn on the human hand gripping an object, for example, a rod (not shown), the palm and the ventral sides of the fingers press against the rod to facilitate the grip,

allowing the outer non-skid layer **23** on the rear sleeve portion **13** and the front sleeve portion **11** to come into contact with the rod to provide a non-skid effect.

[0034] In the first embodiment illustrated by FIG. **1** through FIG. **4**, for example, when the sleeve body **19** is worn on the human hand, with the palm pressing against the ground, the inner non-skid layer **21** and the inner surface **191** of the rear sleeve portion **13** as well as the outer non-skid layer **23** and the outer surface **192** of the rear sleeve portion **13** are necessarily disposed at the sleeve body **19** and correspond in position to a point at which the palm presses against the ground. Therefore, the palm pressing against the ground is accompanied by the non-skid effect regardless of whether the human hand is sweating.

[0035] In the first embodiment illustrated by FIG. **2**, for example, the inner non-skid layer **21** and the outer non-skid layer **23** are disposed at the sleeve body **19** and correspond in position to the back of the human hand. In the disclosure, the back of the human hand includes the backs of the fingers.

[0036] In the first embodiment illustrated by FIG. **1** through FIG. **4**, in addition to the front sleeve portion **11** and the rear sleeve portion **13**, the inner-outer non-skid sleeve **10** further comprises a radical sleeve portion **15**. The radical sleeve portion **15** is fitted to the front sleeve portion **11** and the rear sleeve portion **13**. The opening is formed at the radical sleeve portion **15** and is in communication with the inside of the sleeve body **19**. The radical sleeve portion **15** is fitted around the wrist of the human hand. A surface of the radical sleeve portion **15** is defined as an inner surface **191** of the sleeve body **19** when located inside the sleeve body **19**. Similarly, the radical sleeve portion **15** has the inner non-skid layer **21**, and the outer surface **192** has the outer non-skid layer **23**. Referring to FIG. **5**, the diagram shows the state where the sleeve body **19** does not have the radical sleeve portion, showing a glove adapted to be fitted to three fingers only.

[0037] In the first embodiment illustrated by FIG. **1** through FIG. **4**, for example, the inner non-skid layer **21** and the outer non-skid layer **23** are disposed at the sleeve body **19** and correspond in position to a side position SP of the human hand. The side position SP includes the knifehand position of the human hand and the lateral sides of the fingers of the human hand.

[0038] In the first embodiment illustrated by FIG. **6**, a plurality of penetrating holes **17** penetrate the sleeve body **19**, the inner non-skid layer (not shown) and the outer non-skid layer **23**. External air enters the sleeve body **19** through the plurality of penetrating holes **17**. Air and moisture in the sleeve body **19** escapes through the plurality of penetrating holes **17**, achieving ventilation.

[0039] In the first embodiment, the inner non-skid layer **21** and the outer non-skid layer **23** are strip-shaped and intertwined with each other for an exemplary purpose but may also be in any other shapes, for example, dot-shaped and circle-shaped as shown in FIG. **7** and FIG. **8**, or in any other shapes or irregular shapes, which can be directly understood and thus are, for the sake of brevity, not depicted by the accompanying drawings. As shown in FIG. **9**, the outer non-skid layer **23** and the inner non-skid layer (not shown) are provided in the form of an integrated sheet and thus disposed outside the sleeve body **19** and at the palmar position inside the sleeve body **19**, whereas the outer non-skid layer **23** and the inner non-skid layer (not shown) are not disposed on the lateral sides and rear side of the sleeve body **19**; thus, when the inner-outer non-skid sleeve **10** is worn on the human hand to allow the human hand to be positioned in the sleeve body **19**, the palm of the human hand is still in contact with the inner non-skid layer (not shown), whereas the back of the human hand and the knifehand position are in contact with the inner surfaces **191** of the front sleeve portion **11** and the rear sleeve portion **13**. FIG. **9** depicts a finger-exposed glove, with the front sleeve portion **11** not entirely enclosing the fingers but partially exposing the fingers.

Furthermore, the inner non-skid layer **21** and the outer non-skid layer **23** are made of Silica gel, Polyurethane (PU), rubber or knitted fabric, and exemplified by Silica gel in the first embodiment. As shown in FIG. **10**, the outer non-skid layer **23** is disposed on a pad body **25** and then disposed at the sleeve body **19** to achieve the same aforesaid result-allowing the outer non-skid layer **23** to be disposed at the sleeve body **19**. By doing so, the inner non-skid layer **21** can be positioned in place

and thus is, for the sake of brevity, not depicted by the accompanying drawings. The pad body **25** is made of cloth, leather or any other known, existing materials.

[0040] As shown in FIG. **11**, the sleeve body **19** is also provided in the form of a glove, wherein four fingers other than the thumb are combined. Although the inner non-skid layer **21** (not shown) and the outer non-skid layer **23** are not disposed at the side position SP of the sleeve body **19**, a non-skid effect still occurs on the back of the human hand and when the palm presses against the ground. The inner non-skid layer **21** and the outer non-skid layer **23** disposed at the sleeve body **19** do not necessarily correspond in position to the hand back position, provided that hand back skidding prevention is not required. The aforesaid form can be directly understood and thus is, for the sake of brevity, not depicted by the accompanying drawings. As shown in FIG. **12**, in some extreme examples, the sleeve body **19** provided in the form of a glove has a palm portion but not any finger portion as shown in FIG. **12**. However, the sleeve body **19** in FIG. **12** still has the front sleeve portion **11** and the rear sleeve portion **13**, but the front sleeve portion **11** merely encloses the roots of the fingers without extending by a predetermined length; this, however, still falls within the scope of the claims of the disclosure.

[0041] A conventional knitted fabric has a base and fibers on the surface of the base. When the inner non-skid layer **21** and the outer non-skid layer **23** are positioned in place on the surface of the sleeve body **19**, if they are made of Silica gel or any other materials that have to be heated up and melted, the fibers on the surface of the knitted fabric will curl and adhere to the base of the knitted fabric because of the heat. As a result, the inner non-skid layer **21** and the outer non-skid layer **23** look seemingly lower than the fibers on the surface of the knitted fabric sometimes. In fact, the inner non-skid layer **21** and the outer non-skid layer **23** still adhere to the base of the knitted fabric and protrude from the base of the knitted fabric. The inner non-skid layer **21** and the outer non-skid layer **23** still protrude from the surface of the sleeve body **19** when the sleeve body **19** is a knitted fabric.

[0042] Therefore, the inner-outer non-skid sleeve **10** worn on the human hand absorbs sweat and provides a non-skid effect on the outside and the inside regardless of whether the human hand is sweating.

[0043] As shown in FIG. **13** and FIG. **14**, the second preferred embodiment of the disclosure provides an inner-outer non-skid sleeve **10'** worn on a hand or foot. The second preferred embodiment is distinguished from the first preferred embodiment by the technical features described below.

[0044] The sleeve body **19'** is provided in the form of a sock, with the front sleeve portion **11'** being fitted to the toes of the human foot, and the rear sleeve portion **13'** being fitted to the sole of the human foot.

[0045] The toe prints of the toes of the human foot in the sleeve body **19'** provided in the form of a sock in the second embodiment correspond in position to the fingerprints of the fingers of the human hand in the first embodiment, with the knife foot corresponding in position to the knife hand, the sole corresponding in position to the palm, the dorsal side of the foot corresponding in position to the back of the hand, and the ankle corresponding in position to the wrist. Therefore, the arrangement of the inner non-skid layer **21'** and the outer non-skid layer **23'** is the same as that disclosed in the first embodiment and the diagrams thereof.

[0046] Therefore, although the second embodiment is different from the first embodiment in terms of forms and wearing positions, the second embodiment is effective in absorbing sweat and providing a non-skid effect on the outside and the inside regardless of whether the human foot or hand is sweating.

[0047] The other structural features and achievable advantages of the second embodiment are substantially identical to those of the first embodiment and thus are, for the sake of brevity, not reiterated.

[0048] The disclosure is disclosed above by embodiments. The embodiments are illustrative of the

disclosure but shall not be interpreted as restrictive of the scope of the claims of the disclosure. Thus, all simple variations or equivalent implementation carried out to the aforesaid embodiments according to the claims and detailed description of the disclosure shall be deemed falling within the scope of the claims of the disclosure.

Claims

1. An inner-outer non-skid sleeve worn on a hand or foot, comprising: a front sleeve portion and a rear sleeve portion coupled together, the front sleeve portion being fitted to fingers of a human hand or toes of a human foot, and the rear sleeve portion being fitted to a palm of the human hand or a sole of the human foot, wherein the front sleeve portion and the rear sleeve portion are made of a flexible, hygroscopic material and joined to form a sleeve body and an opening, allowing the human hand or the human foot to be inserted into the sleeve body through the opening, wherein surfaces of the front sleeve portion and the rear sleeve portion are defined as inner surfaces when located inside the sleeve body, and the inner surfaces of the front sleeve portion and the rear sleeve portion have an inner non-skid layer so as to be partially covered with the inner non-skid layer but partially exposed, wherein the human hand or the human foot comes into contact with the inner non-skid layer and the inner surfaces of the front sleeve portion and the rear sleeve portion when the inner-outer non-skid sleeve is worn on the human hand or the human foot to allow the human hand or the human foot to be positioned in the sleeve body, wherein the inner non-skid layer has a predetermined thickness and thus protrudes from the inner surfaces of the sleeve body, wherein surfaces of the front sleeve portion and the rear sleeve portion are defined as outer surfaces when located outside the sleeve body, and the outer surfaces of the front sleeve portion and the rear sleeve portion have an outer non-skid layer so as to be partially covered with the outer non-skid layer but partially exposed, wherein the outer non-skid layer is in contact with a ground when the sleeve body is worn on the human hand or the human foot with the palm pressing against the ground or the human foot treading on the ground, wherein the outer non-skid layer has a predetermined thickness and thus protrudes from the outer surfaces of the sleeve body.
2. The inner-outer non-skid sleeve worn on a hand or foot according to claim 1, wherein the inner non-skid layer and the inner surface of the front sleeve portion are necessarily disposed at the front sleeve portion and correspond in position to fingerprints and ventral sides of the fingers of the human hand or toe prints of the toes of the human foot.
3. The inner-outer non-skid sleeve worn on a hand or foot according to claim 1, wherein the outer non-skid layer and the outer surface of the front sleeve portion are necessarily disposed at the front sleeve portion and correspond in position to fingerprints and ventral sides of the fingers of the human hand or toe prints of the toes of the human foot.
4. The inner-outer non-skid sleeve worn on a hand or foot according to claim 1, wherein when the sleeve body is worn on the human hand or the human foot with the palm pressing against the ground or the human foot treading on the ground, the inner non-skid layer and the inner surface of the rear sleeve portion as well as the outer non-skid layer and the outer surface of the rear sleeve portion are disposed at a ground-pressing force acting point of the sleeve body.
5. The inner-outer non-skid sleeve worn on a hand or foot according to claim 1, wherein the inner non-skid layer and the outer non-skid layer are disposed at the sleeve body formed by joining the front sleeve portion and the rear sleeve portion and correspond in position to the back of the human hand or the dorsal side of the human foot.
6. The inner-outer non-skid sleeve worn on a hand or foot according to claim 1, wherein the sleeve body further comprises a radical sleeve portion coupled to the front sleeve portion and the rear sleeve portion, with the opening formed at the radical sleeve portion and being in communication with the inside of the sleeve body, and the radical sleeve portion is fitted around a wrist of the human hand or an ankle of the human foot.

7. The inner-outer non-skid sleeve worn on a hand or foot according to claim 6, wherein the inner non-skid layer is disposed on an inner surface of the radical sleeve portion, and the outer non-skid layer is disposed on an outer surface of the radical sleeve portion.

8. The inner-outer non-skid sleeve worn on a hand or foot according to claim 1, wherein the inner non-skid layer and the outer non-skid layer are disposed at the sleeve body and correspond in position to a lateral side of the human hand or the human foot when the sleeve body is worn on the human hand or the human foot.

9. The inner-outer non-skid sleeve worn on a hand or foot according to claim 8, wherein the inner non-skid layer and the outer non-skid layer are disposed at the sleeve body and correspond in position to a knifehand of the human hand or a knifefoot of the human foot when the sleeve body is worn on the human hand or the human foot.

10. The inner-outer non-skid sleeve worn on a hand or foot according to claim 1, further comprising a plurality of penetrating holes penetrating the sleeve body, the inner non-skid layer and the outer non-skid layer.
